Description	Company	Model	Unit $	#	Total $
Breakout board					
Complete assembly			100	1.00	100.00
3D printed housing				1.00	0.00
Power supply					
Power adapter 3 V	NC	BY0301&BK	5.57	1.00	5.57
Power adapter 5 V	MTYTOT	ALT-0503	6.99	1.00	6.99
Power adapter 12 V	SmoTecQ	STQ-12V-2A	5.95	1.00	5.95
3D printed housing				1.00	0.00
Dual-lick detector					
Complete assembly			100	1.00	100.00
3D printed housing				1.00	0.00
Water reservoir, solenoids, and spouts					
Syringe 5 ml	BD Medical	309646	0.18	2.00	0.36
Syringe 30 ml	BD Medical	302832	0.32	2.00	0.64
3-way valve	Cole Parmer	48523-38	1.005	2.00	2.01
Solenoid	The Lee Company	LHQA0531220H	65.6	2.00	131.20
Tubing (ID 1/16, OD 1/8, wall 1/32)	Tygon	E-3603	0.28	10.00	2.80
Stainless steel tube	McMaster-Carr	5560K72	3.9	1.00	3.90
Pedestal post holder	Thorlabs	PH1	8.74	1.00	8.74
Optical post	Thorlabs	TR6	8.66	1.00	8.66
Cage rods	Thorlabs	ER3-P4	7.255	2.00	14.51
Swivel post	Thorlabs	MSWC	40.8	1.00	40.80
Laser cut holder for solenoids				1.00	0.00
3D-printed holder for spouts				1.00	0.00
Head restriction set-up					
Kinematic magnetic base	Newport corporation	BK-3A	159.64	1.00	159.64
Pedestal post holder	Thorlabs	PH1	8.74	1.00	8.74
Optical post	Thorlabs	TR6	8.66	1.00	8.66
Adjustable-Height Optics Clamp	Thorlabs	VG100	93.34	1.00	93.34
Aluminum tube	McMaster-Carr	6061	6.75	1.00	6.75
Banana socket	na	na	1.28	1.00	1.28
Head-restriction plate behavior	SendCutSend	Titanium	1.4	1.00	1.40
Head-restriction plate imaging	SendCutSend	Titanium	1.61	1.00	1.61
Imaging cone	eMachineShop	Aluminum 6061	4.5	1.00	4.50
Magnets plate	K&J Magnetics	1/16" x 1/32"	0.14	3.00	0.42
Magnets cone	The Magnet Baron	2 mm x 0.5 mm	0.15	3.00	0.45
CNC head-restriction holders				2.00	0.00
Breadboard lasercut				1.00	0.00
Motorised manipulator					
Linear track NEMA11 motor (50 mm)	Befenybay	BE073-1	47.3	3.00	141.90
3D-printed optical post holder				1.00	0.00
3D-printed breadboard to linear track				1.00	0.00
Speaker and Amplifier					

Amplifiers	DROK	DROK	10.75	1.00	10.75
Speakers	Intervox	S150RMSS-R2W	0.75	2.00	1.50
L-Bracket	McMaster-Carr	1556A54	1.06	1.00	1.06
Optical Post	Thorlabs	TR3	6.58	2.00	13.16
3D-printed casing for speaker-amplifier				1.00	0.00
Stepper-motor drivers					
Stepper-motor drivers	Banria	4257	26.9	3.00	80.70
Power supply	DROK	24V	15.89	1.00	15.89
Stepper motor housing (laser cut)				1.00	0.00
Optogenetics-power modulation					
Complete assembly			25	1.00	25.00
3D printed housing				1.00	0.00
Miscellaneous					
4-40 Screws	Thorlabs	SH4S025	0.18	10.00	1.80
1/4" Screws	Thorlabs	SH25S025	0.55	10.00	5.50
Table Clamp	Thorlabs	CL5	5.67	2.00	11.34
BNC cables	50 ohm	3 feet	5.63	5.00	28.15
22 AWG wiring	Tensility	30-00689	2	10.00	20.00
Banana plug	na	na	1.7	1.00	1.70
Computer	Dell	Optiplex 7060	276	1.00	276.00
Screen	Dell	SE2422H 24"	90	1.00	90.00
12" x 18" plate				1.00	0.00
Optional					
Breadboard 12" x 18" x 1/2"	Thorlabs	MB1218	459.36	1.00	459.36
Breadboard 6" round x 1/2"	Thorlabs MBR6	MBR6	156.35	1.00	156.35
Total costs rig					**1077.37**
Including computer					**1443.37**
Including optional					**2059.08**